



Software Testing

Chapter 4

TEST IMPLEMENTATION AND TEST REPORT

Session 2

TEST REPORT



Contents



Test Report



Defect Management



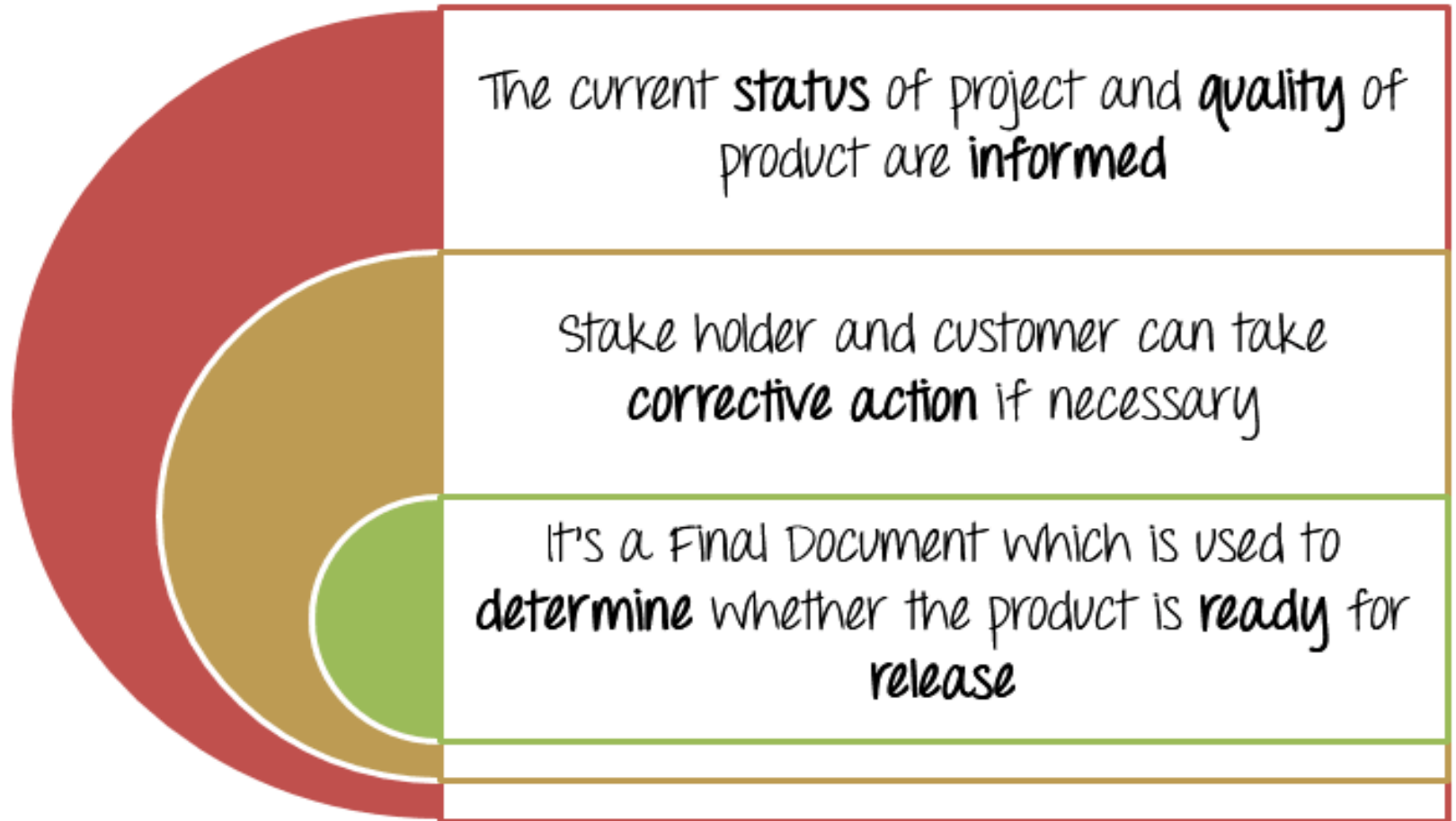
1. Test Report

- ❖ It is a document which contains a summary of all test activities and final test results of a testing project.
- ❖ Test report is an assessment of how well the Testing is performed.
- ❖ Based on the test report, stakeholders can evaluate the quality of the tested product and make a decision on the software release



1. Test Report

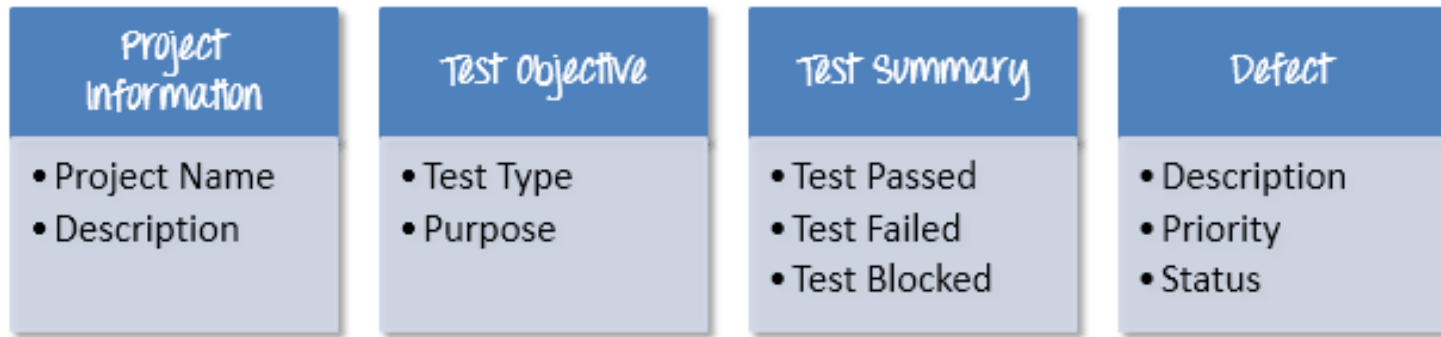
❖ Benefits of test report





1. Test Report

❖ Test report contains:



❖ A good test report should be

- Detail
- Clear
- Standard
- Specific





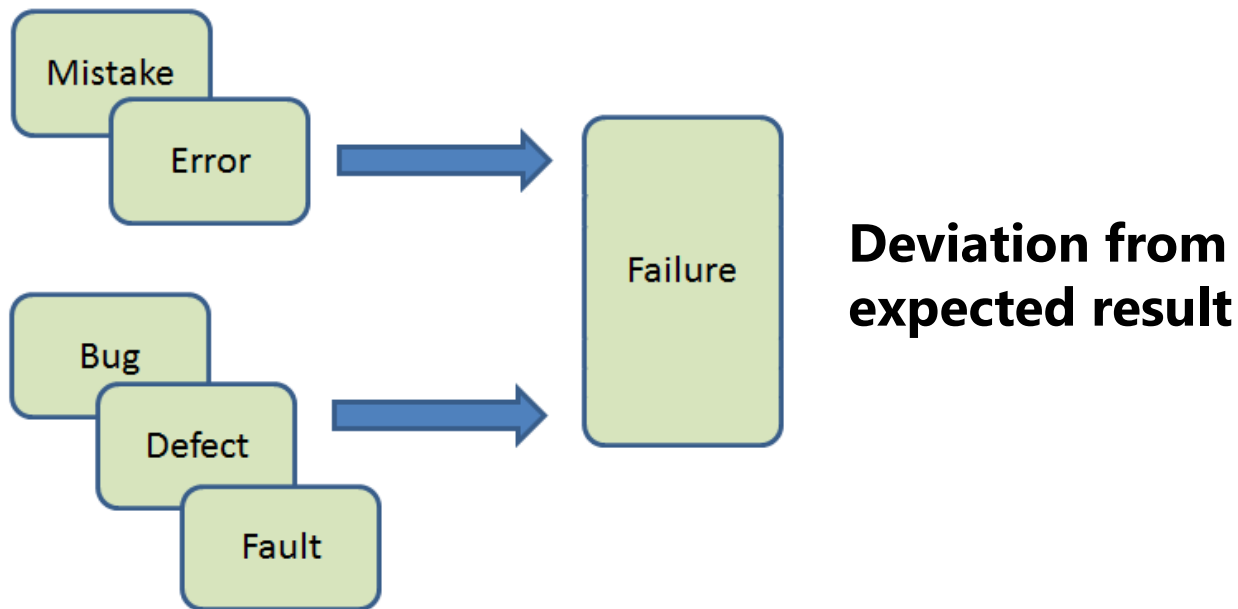
2. Defect Management

- ❖ Defect Life cycle
- ❖ Defect handling process
- ❖ Priority and Severity of bug
- ❖ Bug report



What is a bug?

- ❖ A bug is known as defect or fault which is a manifestation of an error in software.





What is a bug?

❖ Bug types:

- Bugs in requirements
 - ✓ When requirement is incorrect, unambiguous, inconsistent or incomplete
 - ✓ When requirement contradict to business flow in case when it exists
- Functional bugs in application
 - ✓ Nonconformance to requirements
 - ✓ Something that does not correspond to valid Customer's expectations/common sense that are assumed but may be not described in product requirements.

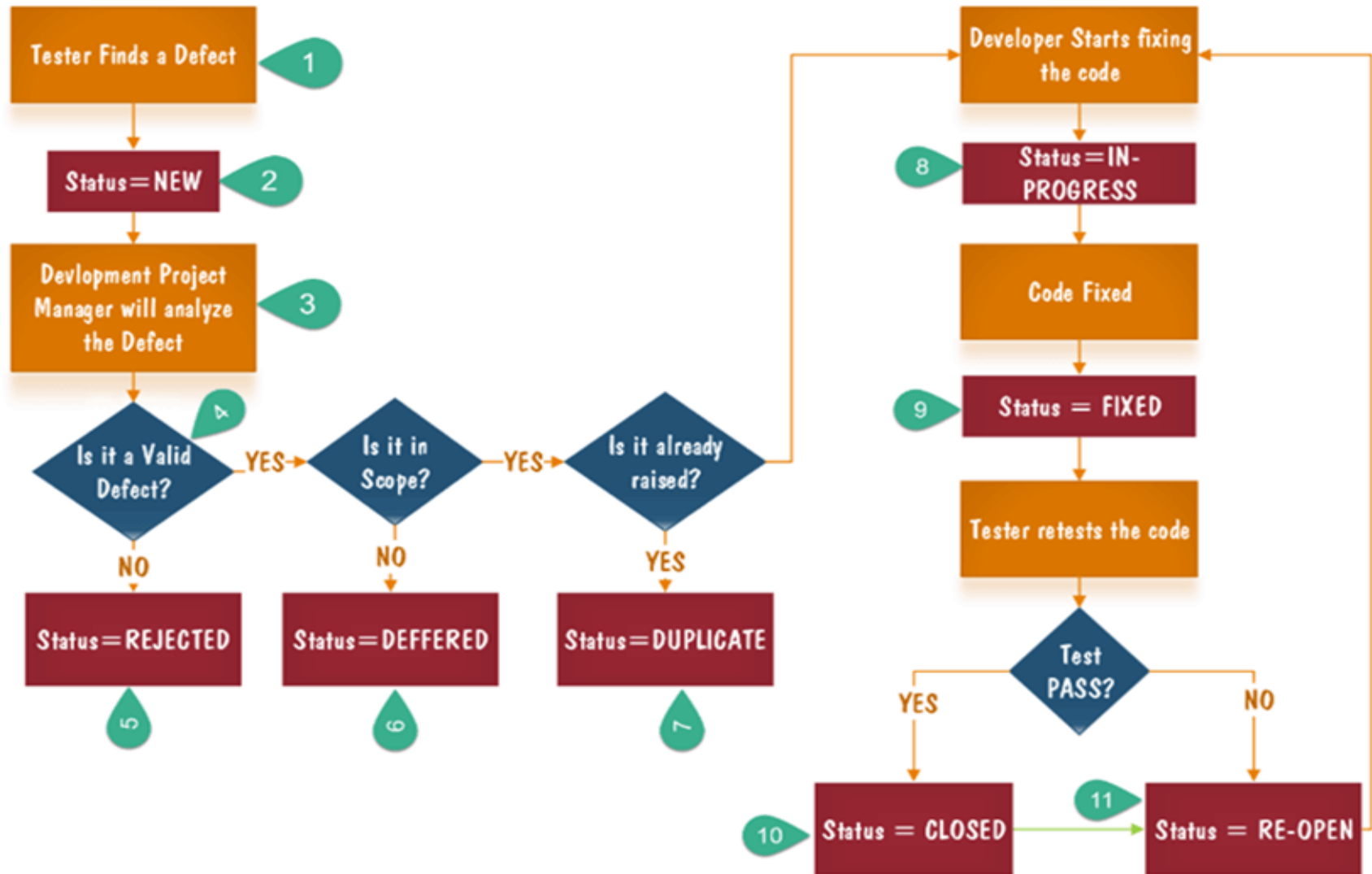


2.1. Defect Life cycle

- ❖ Defect life cycle is a cycle which a defect goes through during its lifetime.
- ❖ It starts when defect is found and ends when a defect is closed, after ensuring it's not reproduced.
- ❖ Defect life cycle is related to the bug found during testing.
- ❖ The bug has different states in the Life Cycle: New, rejected, deferred, duplicate, in progress, fixed, re-open, closed



2.1. Defect Life Cycle





2.1. Defect Life Cycle

1. Tester finds the defect
2. **New:** When a defect is logged and posted for the first time. It's state is given as new.
3. Defect is forwarded to Project Manager for analyze
4. Project Manager decides whether defect is valid
5. Here the defect is not valid- status given "Rejected."
6. If the defect is not rejected then the next step is to check whether it is in scope.
7. Next, manager verifies whether a similar defect was raised earlier. If yes defect is assigned a status duplicate.



2.1. Defect Life Cycle

8. If no the defect is assigned to the developer who starts fixing the code. During this stage, the defect is assigned a status in- progress.
9. Once the code is fixed. Defect is assigned a status fixed
10. Next the tester will re-test the code. In case, the Test Case passes the defect is closed.
11. If the test cases fails again, the defect is re-opened and assigned to the developer.



2.2 Defect Handling Process

❖ The defect management process includes several stages, which are as follows:

Defect Management Process





2.2 Defect Handling Process

1. Defect Prevention:

In this stage, the execution of procedures, methodology, and standard approaches decreases the risk of defects.



Defect Prevention

01

Estimate Predictable
impact

02

Minimize expected
impact

03

Identify Critical Risk



2.2 Defect Handling Process

2. Deliverable Baseline

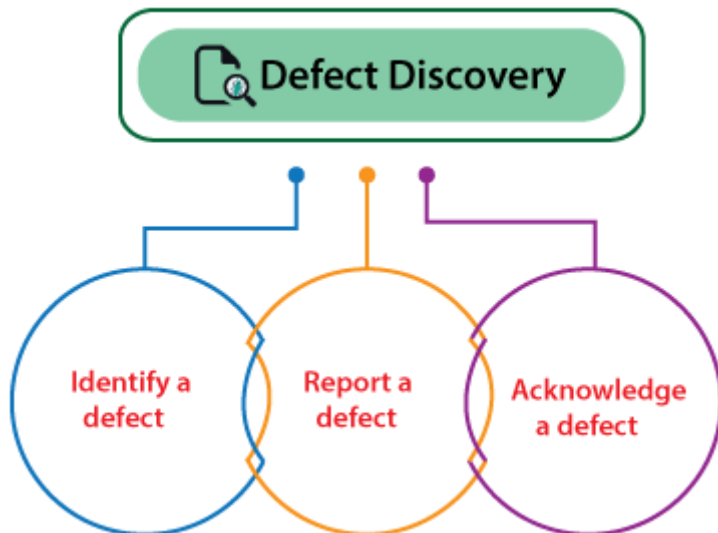
- ❖ The **deliverable is a baseline** as soon as a deliverable reaches its pre-defined milestone.
- ❖ In this stage, the deliverable is carried from one step to another; the system's existing defects also move forward to the next step or a milestone.
- ❖ The pre-defined milestone describes what software is supposed to accomplish.



2.2 Defect Handling Process

3. Defect Discovery

If developers have approved or documented the defect as a valid one, then only a defect is considered as ***discovered***.





2.2 Defect Handling Process

4. Defect Resolution

This process is used to specified and track the defects.





2.2 Defect Handling Process

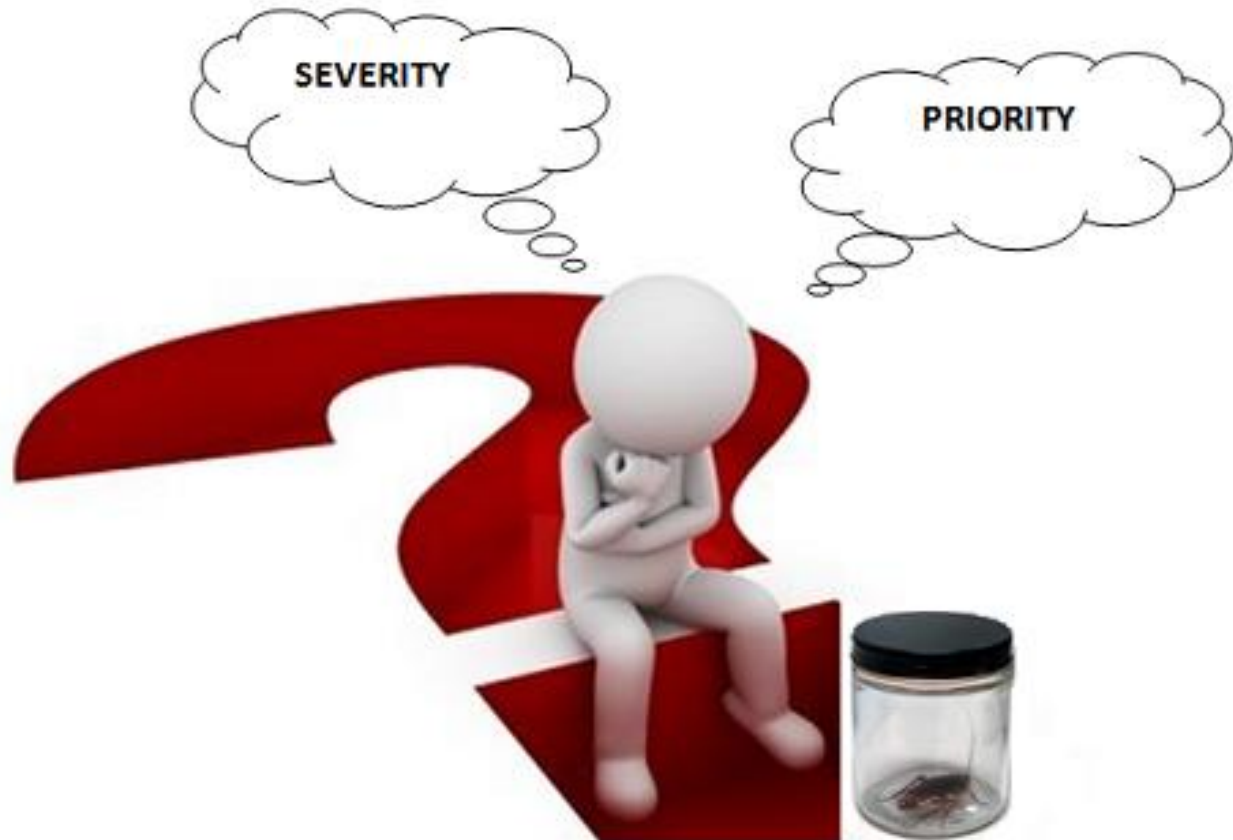
5. Process Improvement

- ❖ In this phase, we will look into the lower priority defects because these defects are also essential as well as impact the system.

6. Management Reporting

- ❖ The management reporting is required to make sure that the generated reports have an objective and increase the defect management process.

2.3. Priority and Severity





Severity defines the impact that a given defect has on the system, so how bad the defect is.



- Impact on functionality
- How serious is the bug?

Priority is defined as the order in which a defect should be fixed in accordance to business value or other project/customer needs



- Scheduling
- How soon to fix the bug?

Severity ↑	High Severity, Low Priority 3	High Severity, High Priority 1
	Low Severity, Low Priority 4	Low Severity, High Priority 2
	Priority →	



Severity levels

Critical

- The failed function is unusable and there is no acceptable alternative method to achieve the required results.

Major

- The failed function is unusable but there exists an acceptable alternative method to achieve the required results.

Medium

- The defect that does not result in the termination, but causes the system to produce incorrect, incomplete or inconsistent results.

Minor

- The defect that does not result in the termination and does not damage the usability of the system and the desired results can be easily obtained by working around.



Priority levels

Critical

- should be resolved immediately

High

- should be resolved as soon as possible in the normal course of development activity, before the software is released

Medium

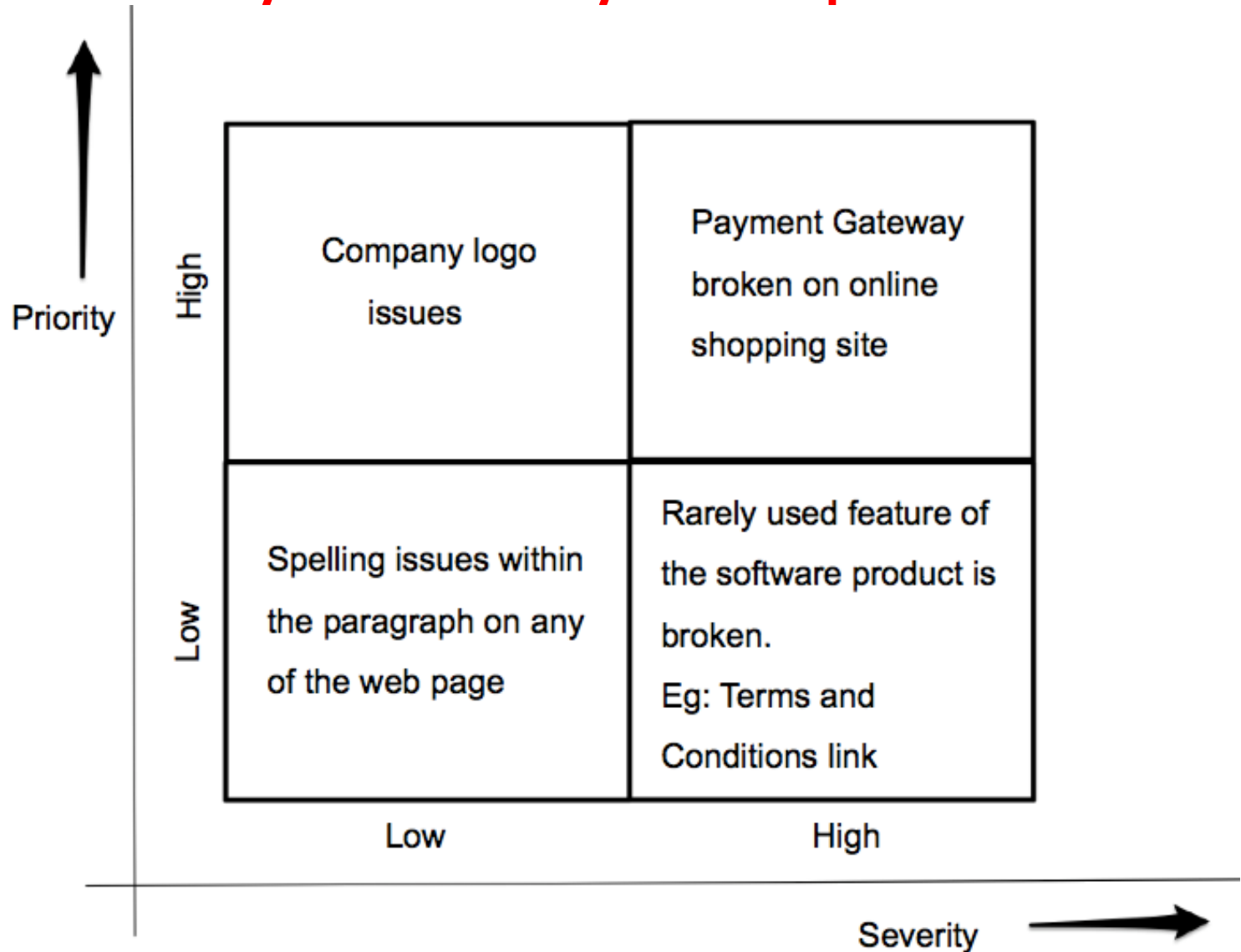
- should be repaired after serious bugs have been fixed

Low

- can be resolved in a future major system revision or not be resolved at all



Severity & Priority examples





2.4. Bug Report

❖ **Defect report** is a technical document written to describe the symptoms of a bug to:

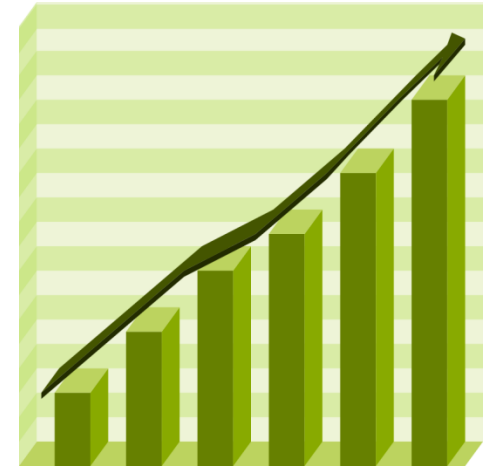
- communicate the impact of a quality problem
- prioritize the bug for fixing
- help the programmer locate the underlying defect and fix it





2.4 Bug Report

❖ Why do we need to report bugs?



- 1 Do not forget to fix and retest all found issues
- 2 Measure product quality
- 3 Define risky areas/configurations/testing types/etc.



2.4 Bug Report

❖ Bug report provides information to

Tester

- How to reproduce?
- Why is it a bug?
- Expected result

Developer

- How to reproduce?
- Why is it a bug?
- What is expected result and why?

Customer

- How bad is the bug?
- Affected area



2.4 Bug Report

- ❖ Bug report can contains sets of mandatory and optional fields
- ❖ What fields are mandatory and what are optional and filling rules are defined on each project
- ❖ Most bug tracking system allow to configure these fields and rules





2.4 Bug report: mandatory fields

Mandatory bug report fields:

- Summary
- Severity
- Priority
- Description
- Steps to Reproduce
- Actual and Expected Results
- Attachment

The screenshot shows a bug report form with the following fields:

- Summary:** Special Location field doesn't become editable after Company Code selecting
- Priority:** Very High (with a dropdown arrow and a help icon)
- Assignee:** Voronetski, Maxim (with a dropdown arrow)
- Description:**
 - "Special Location" field is only available for editing after selecting appropriate "Company Code" and clicking "Refresh Rates & Employee Data" button.
 - *Steps to reproduce:***
 1. Open Workbook that is not in read-only status.
 2. Create new resource.
 3. Type in resource's name.
 3. Select Resource Type = "Internal Resource".
 4. Select Company Code = "0008 (SAP AMERICA, INC.)".
 - *Expected result:***
Special Location field becomes editable.
 - *Actual result:***
Special Location field remains disabled (see attachment-1) until "Refresh Rates & Employee Data" button is clicked (see attachment-2).
- Attachment:** Drop files to attach, or [browse](#).

At the bottom right, there are buttons for **Update** and **Cancel**.



2.4 Bug report: Summary

Summary gives **enough information to understand what the problem is and how you can reproduce it.**

Good summary is built based on the following schema:

What – Where – When

Example: Bad summary:

It's impossible to save Concept

Example: Good summary:

It's impossible to save Concept with long description created via HTML Editor



2.4 Bug report: Description

- ❖ Description gives detailed and clear interpretation of a problem, explains why it is a bug, includes the following information:
 - Description of application behavior (e.g. interpretation of test failures)
 - Justifications of why this is a bug
 - Any relevant links
 - Interpretation of the specification
 - Examples from other areas

Not copy of summary!



2.4 Bug report: Steps to reproduce

❖ General recommendations:

- Show the scenario how to recreate the bug on any system
- Check that steps provide clear instructions, so that others can consistently reproduce it
- Use the rule: new action = new step
- Include test data on which is reproduced
- Test your own instructions

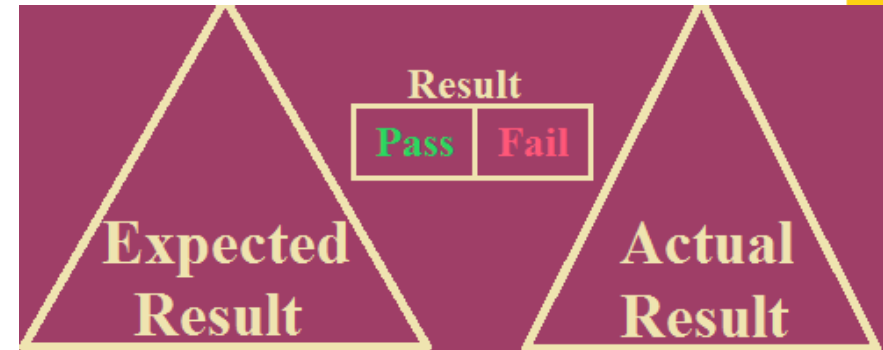


2.4 Bug report: Actual & Expected results

The test results, including Expected Result and Actual Result, will show what's wrong.

Actual Result describes what actually happened, what currently happens when the bug is present.

Expected Result describes what should happen if the bug was fixed.



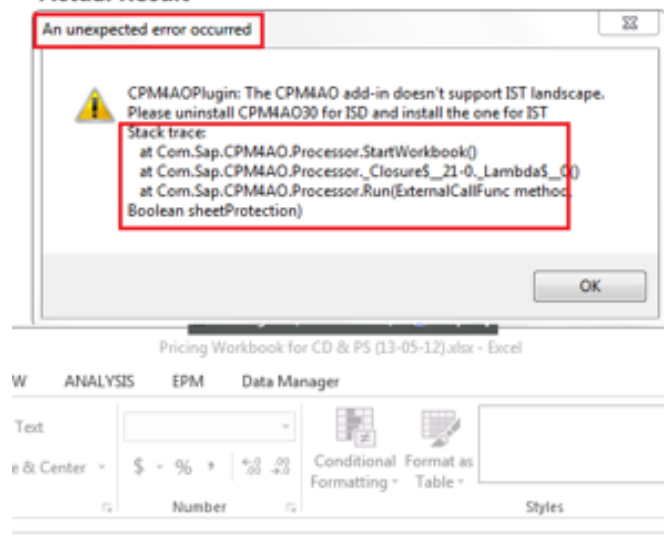


2.4 Bug report: Attachment

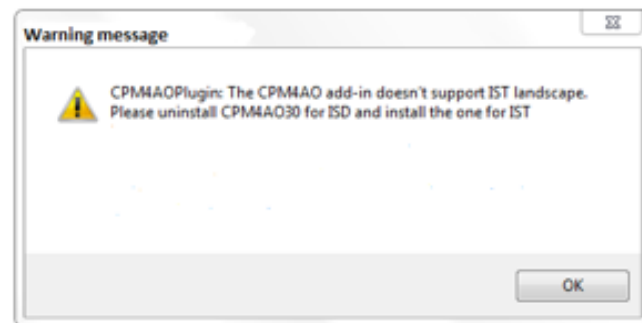
Attachment can be like:

- Pictures (screenshots)
- Files (any kind of logs)
- **DB query (to get test data);**
- Documents
- **Video (just in case of hard reproducible bug)**

Actual Result



Expected result





2.4 Bug Report: Optional bug report fields

❖ List of the most common and useful optional fields:

- Issue Type:
 - Regression issues/New Feature/Configuration/Integration/etc.
- Component:
 - Functional components/areas in application, e.g. payments, user login
- Environments:
 - QA, DEV, UAT, PROD
- Found Version
- Fix Version
- Labels
- ...



2.4 Bug report: A good defect example

Summary Special Location field doesn't become editable after selecting Company Code

Severity Major

Priority High

Steps to reproduce "Special Location" field is only available for editing after selecting appropriate "Company Code" and clicking "Refresh Rates & Employee Data" button

1. Open Workbook that is not in read-only status
2. Create new resource
3. Type in resource's name
4. Select one of Resource Type, for example "Internal Resource"
5. Select appropriate Company Code, "0008" for our example

Expected Results Special Location field becomes editable

Special Location field remains disable (see attachment-1) until "Refresh Rates &

Actual Results Employee Data" button is clicked (see attachment-2)



2.4 Bug report: A good defect example



IT M-827

Special Location field doesn't become editable after Company Code selecting



Edit



Comment

Assign

More ▾

Reopen

Blocked

Workflow ▾

Details

Type:



Bug

Priority:



Very High

Component/s:

IT_Commercial Project

Labels:

AO.Plugin

VB.Net

Environment:

IST, ISD

Epic Link:

Test Findings

Sprint:

Sprint27 (CW5-7)

Rank:

0|i5csrb:

Status:

DONE

(View Workflow)

Resolution:

Done

Version / Delivery:

RD2017.02

Description

"Special Location" field is only available for editing after selecting appropriate "Company Code" and clicking "Refresh Rates & Employee Data" button.

Steps to reproduce:

1. Open Workbook that is not in read-only status.
2. Create new resource.
3. Type in resource's name.
3. Select Resource Type = "Internal Resource".
4. Select Company Code = "0008 (SAP AMERICA, INC.)".

Expected result:

Special Location field becomes editable.

Actual result:

Special Location field remains disabled (see attachment-1) until "Refresh Rates & Employee Data" button is clicked (see attachment-2).



What is a bad bug report?

Bad bug report is a bug report that duplicates already existing bugs

T	P	Key	Summary
		QP-3108	[Quality\Checksheet] Error on opening "Failure Mode" picker
		QP-1602	Checksheets: 'Failure Mode Picker List' picker opening causing the exception
		CR-1827	Checksheets - Failure Mode Picker Error

Bad bug report is a bug report that contains just opinions or complaints

Examples:

- It doesn't work!
- My computer crashed!
- ...



What is a bad bug report?

Bad bug report is a bug report that doesn't clearly specify the problem

Heat map view highlighting is not correct

Description

1. log in
2. switch to Heat Map
3. hover over a customer (or different customers)

no any information about problem source anywhere

Actual result: highlighting is not correct

Expected result: highlighting must be correct



What is a bad bug report?

Bad bug report is a bug report that contains grammar or spelling mistakes, or written using not appropriate language

Add Exception function not work

Environment: IE 11; Chrom; FireFox

Description

Add Exception function does not work.

Preconditions:

- Loading Exceptions Use setting should be checked.
- Error Type field should be filtered by 'Customer Part not Setup' or 'Customer Part/Rev not Setup' values.
- Search result appear

Steps:

1. Mark any record from search results table
2. Click **Add Exception** button
3. In pop-up window type any text in Note field
4. Click **OK** button

Expected result: Pop-up window is closed. Exception added.

Actual result: Pop-up window is not closed. Error message appear.



What is a bad bug report?

Bad bug report is a bug report that contains several issues

Exception! Many UI issues on 1 page - can be reported in one bug report

Inventory column incorrect behavior.

Description

Issue 1

Incorrect coloring when Inventory Display setting = 1

Act res: picture 1

Exp res: picture 2

Issue 2

Incorrect column value in case when

Inventory Display setting = 1

and

Units of Measure Select Display setting = FALSE

Act res: picture 3

Exp res: picture 4

Issue 3

Text coloring should depend on Produced Display setting in case when Inventory Display setting > 1.

If Produced Display setting = FALSE:

Act res: picture 5

Exp res: picture 6



ĐẠI HỌC ĐÀ NẴNG
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Vietnam - Korea University of Information and Communication Technology

Thank You !